iSCSI

Introduction

- iSCSI (Internet Small Computer System Interface) is a TCP/IP-based storage networking standard for connecting data storage services
- It provides an inexpensive and easy-to-implement block-level access to remote storage by transporting SCSI commands over a standard Ethernet network
- Effectively, iSCSI emulates the high-performance local storage bus subsystem and creates a logical Storage Area Network (SAN)
- **Key Characteristics**
  - iSCSI requires no specialized cabling or connectivity devices
  - It runs over existing switching and IP infrastructure
  - **Default TCP port**: iSCSI uses **TCP port 3260** by default
  - iSCSI provides **block-level access**, not file-level access
  - Windows Server natively includes both the iSCSI Target Server role and the iSCSI Initiator, with no additional licensing cost

iSCSI Architecture and Components



- IP Network
  - Standard Ethernet adapters and switches connect to iSCSI storage devices.
  - The network should provide at least 1 Gbps throughput and include multiple physical paths to the iSCSI target
- iSCSI Target → Server
  - Allows access to its storage over an IP network
  - Conceptually equivalent to a SCSI controller providing access to local storage.
  - Windows Server provides the iSCSI Target Server as a role service of the File and Storage Services role
- iSCSI Initiator → client
  - Client-side component that provides connectivity to a remote iSCSI target
  - The Windows operating system includes the iSCSI initiator software by default
- IQN (iSCSI Qualified Name)
  - Unique identifiers that iSCSI uses to designate and locate initiators and targets on an iSCSI network
  - When configuring an iSCSI target, you must configure the IQN for the iSCSI initiators that will connect
- iSNS (Internet Storage Name Service)
  - Facilitates discovery, management, and configuration of iSCSI targets

iSCSI Functionality

- **iSCSI Target Server Role Service**
  - **Authentication:** Implements CHAP (Challenge Handshake Authentication Protocol) to authenticate initiator connections, and reverse CHAP to enable the initiator to authenticate the iSCSI target
  - **Virtual Disks:** Implements iSCSI virtual disks as **VHD or VHDX files**, with VHDX supporting up to **64 TB** capacity
  - **Management:** Supports management via Server Manager, Windows PowerShell, and System Center Virtual Machine Manager
  - **Scale Limits:** Scales to **256 iSCSI targets** and **512 virtual disks** per server
- **iSNS Role Service**
  - Hosts a database of active iSCSI nodes (initiators, targets, and management systems)
  - Supports automatic registration of initiators and targets, allowing initiators to query the iSNS server for available targets
  - Implements a State Change Notification Service that triggers notifications to registered clients when database changes occur
  - Offers a Discovery Domain Service that restricts connectivity between designated initiators and targets by grouping them into discovery domains

Deployment Considerations



- **Network Design**
  - When implementing iSCSI networking, you should use a **separate, dedicated network** or configure **Quality of Service (QoS)** on your existing network
  - To ensure sufficient performance, the network should provide at least 1 Gbps throughput and include multiple physical paths to the iSCSI target
- **High Availability**
  - For highly available storage, you can choose between Fibre Channel storage, a SCSI storage array, or a **Storage Spaces Direct cluster**, with the iSCSI Target Server configured as a **cluster role**
- **Network Adapter Considerations**
  - Network adapters with hardware offload capabilities such as **large send/TCP segmentation offload** and **checksum offload** are recommended
  - These features move processing from the CPU to the adapter, freeing the processor for other tasks
  - A Gigabit or 10 GB network provides low latency while maximizing throughput